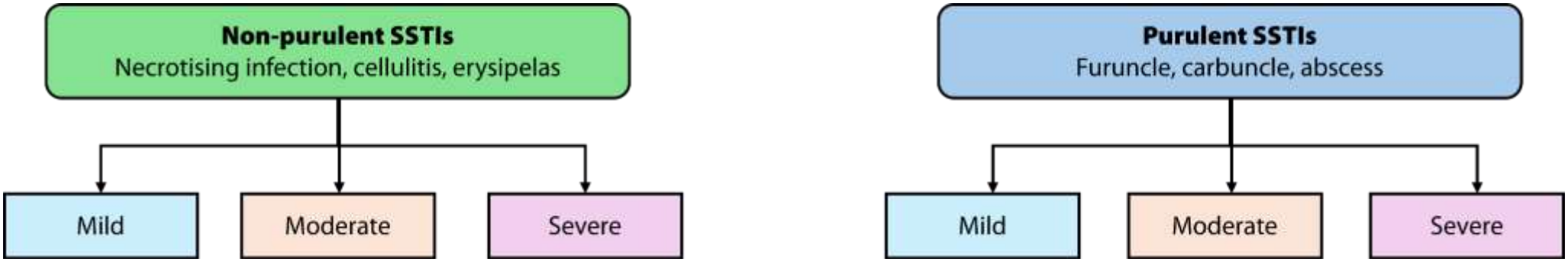
3.10 Skin and Soft Tissue Infections

ADULTS



Ref: *Stevens et al. Clin Infect Dis 2014; 59 (2): e10*

Purulent SSTI	
Mild	Localised infection. Incision and drainage indicated. Oral antibiotic can be used if size of the abscess is <5 cm or surrounding erythema is <5cm.
Moderate	Evidence of systemic signs of infection, or extensive lesions. Incision and drainage indicated. Oral or systemic (IV) antibiotics can be used.
Severe	Evidence of severe sepsis or septic shock OR immunocompromised. Incision and drainage indicated. IV antibiotics should be used.

Non-purulent SSTI	
Mild	Localised infection with no evidence of purulence. Oral antibiotics can be used.
Moderate	Evidence of systemic signs of infection, or extensive lesions. Oral or IV antibiotics can be used.
Severe	Evidence of severe sepsis or septic shock, clinical signs of deeper infection, organ dysfunction OR immunocompromised. IV antibiotics should be used.

1. Consider IV antibiotics if there are multiple abscesses or extremes of age or lack of response to incision and drainage alone.
2. Antibiotic allergy refers to a true hypersensitivity reaction (immediate or delayed). Gastrointestinal symptoms are usually side effects and NOT a true allergy.

Section A	Adults

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Purulent SSTI			
Impetigo *Staphylococcus aureus, Streptococcus pyogenes*			

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Localised/Mild	**Topical Treatment:** Topical Mupirocin 2%* q8H for 5-7 days	**Topical Treatment:** Topical Fusidic Acid 2%* q8 - 12H for 5-7 days	• Incision and drainage indicated. • Send pus / exudate for C&S. • De-escalate/ streamline once C&S result is available. • **If oral antibiotics deemed desirable or topical options not available, consider either of the following:** • Cloxacillin 500 mg PO q6H for 5-7 days • Phenoxymethylpenicillin 250 - 500 mg PO q6H for 5-7 days • Cefalexin 500 mg PO q6h for 5-7 days • **Penicillin Allergy:** • Erythromycin Ethyl Succinate 400 mg PO q6H • **If MRSA is suspected:** • Clindamycin 300 mg PO q6H **OR** 450 mg PO q8H • Co-trimoxazole** 480 mg 2 - 4 tablets q12H • *Not available in UMMC formulary. • **Co-trimoxazole may not cover Streptococci.

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Generalised/Moderate	Cloxacillin 500 mg PO q6H	Cefalexin 500 mg PO q6H	• Duration: 7 days.
Abscess, Carbuncles & Furuncles *Staphylococcus aureus, Streptococcus pyogenes,* polymicrobial (regional skin flora)			
Mild	Cloxacillin 500 mg PO q6H x 5 days	Cefalexin 500 mg PO q6H x 5 days	• Incision and drainage indicated. Send pus / exudate for C&S. • De-escalate/streamline once C&S result is available. IV antibiotics should be de-escalated to oral once clinical improvement occurs • Total duration 5 - 7 days • Duration is suggested and can be adjusted depending on clinical progress. • For patients with diabetes mellitus or immunocompromised/ in shock / not responding to Co-amoxiclav or Ampicillin/Sulbactam, consider Piperacillin/Tazobactam 4.5 g IV q6H. • For patients with risk for ESBL or failed Piperacillin/ Tazobactam or penicillin allergy, consider Meropenem 1 g IV q8H. • **Penicillin allergy, consider either of the following:**
Moderate	Cloxacillin 2 g IV q6H	Cefazolin 2 g IV q8H	
Severe	Co-amoxiclav 1.2 gm IV q8H	Ampicillin/Sulbactam 3 g IV q6H	

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• Clindamycin 300 mg PO /IV q6H • Co-trimoxazole 480 mg PO 3 - 4 tablets q12H • **If MRSA suspected, consider either of the following:** • Clindamycin 300 mg PO q6H • Co-trimoxazole 480 mg PO 3 - 4 tablets q12H • Vancomycin 20 mg/kg (max 2 g) IV loading dose, followed by 15 mg/kg IV q8 - 12H. • Optimise Vancomycin dose by doing therapeutic drug monitoring. • Note: Co-trimoxazole may not cover Streptococci.
Erysipelas *Streptococcus pyogenes, Staphylococcus aureus*			
Localised/Mild	Cloxacillin 500 mg PO q6H **OR** Phenoxymethylpenicillin 500 mg PO q6H **OR**	Cefalexin 500 mg PO q6H	• Total duration: 5 days • Duration is suggested and can be adjusted depending on clinical progress. • IV antibiotics should be de-escalated to oral once clinical improvement occurs • **If Penicillin allergy, consider either of the following:**

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Amoxicillin* 500 mg PO q6H		• Erythromycin Ethyl Succinate 400 mg PO q6H • Clindamycin 300 mg PO q6H • Clindamycin 600 mg IV q8H ***(for severe infection)*** • Co-trimoxazole 480 mg 3 - 4 tablets PO q12H • Co-trimoxazole 15 mg/kg/day IV (Trimethoprim component) divided into twice dosing ***(for severe infection)***
Systemic	Cloxacillin 2 g IV q6H	Cefazolin 1 g IV q8H	• *If only *S. pyogenes* is strongly suspected or cultured. • Note: Co-trimoxazole may not cover Streptococci.
Cellulitis *Streptococcus pyogenes, Staphylococcus aureus*			
Mild	Cloxacillin 500 mg PO q6H **OR** Phenoxymethylpenicillin * 500 mg PO q6H **OR** Amoxicillin* 500 mg PO q6H	Cefalexin 500 mg PO q6H	• Total duration: 5 - 7 days • Duration is suggested and can be adjusted depending on clinical progress. • Send blood C&S if moderate to severe infection. • De-escalate/ streamline once C&S result is available. • IV antibiotics should be de-escalated to oral once clinical improvement occurs. • Elevation of the leg is important to aid healing.

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Moderate	Cloxacillin 2 g IV q6H	Cefazolin 2 g IV q8H	• *If only *S. pyogenes* is strongly suspected • **If not responded to Cefazolin or Cloxacillin. • For patients with diabetes mellitus or immunocompromised/ in shock / not responding to Co-amoxiclav or Ampicillin/Sulbactam, consider Piperacillin/Tazobactam 4.5 g IV q6H. • For patients with risk for ESBL or failed Piperacillin/Tazobactam or penicillin allergy, consider Meropenem 1 g IV q8H. • ^**Risk for acquiring MRSA infections:** • Had previous MRSA infection / colonisation • History of recent hospitalisation • Recent surgical procedure (especially surgical implant related surgery) • Illicit drug user • Athlete^^ • Men having sex with men (MSM)^^ • Penetration injury • Severe sepsis / shock • Failed initial antibiotic • Impaired host defences • Optimise Vancomycin dose by doing therapeutic drug
Severe If community-acquired MRSA is suspected^^ **OR** If MRSA is suspected in mild infection	Cloxacillin 2 g IV q6H Clindamycin 300 mg PO q6H for 5-7 days	Cefazolin 2 g IV q8H **OR** Co-amoxiclav 1.2 g IV q8H** **OR** Ampicillin/Sulbactam 1.5 g IV q8H** Co-trimoxazole (480 mg tablet) 2-4 tablets PO q12H for 5-7 days	
If MRSA is suspected in severe infection^	Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H		

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			monitoring. • Refer to the ID if allergic to Vancomycin. • Note: Co-trimoxazole does not cover Streptococci. • **Penicillin allergy, consider either of the following:** • Clindamycin 300 mg PO q6H • Clindamycin 600 mg IV q8H ***(for severe infection)*** • Co-trimoxazole 480 mg 2 - 4 tablets PO q12H • Vancomycin 20-25 mg/kg (max 2 g) IV loading dose, followed by 15 mg/kg IV q8-12H • Linezolid 600 mg PO/IV q12H
Diabetic Foot Infection *Streptococcus pyogenes, Staphylococcus aureus,* Gram-negative bacteria, anaerobes			
Mild	Co-amoxiclav 625 mg PO q8H **OR** Co-amoxiclav 1 g PO q12H	Cefalexin 500 mg PO q6H + Metronidazole 400 mg PO q8H	• Send tissue/deep swabs for C&S. • Total duration: 5 - 7 days. Duration is suggested and can be adjusted depending on clinical progress. • IV antibiotics should be de-escalated to oral once clinical improvement occurs. De-escalate /streamline once C&S result is available.

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• If penicillin immediate hypersensitivity, call ID.
Moderate	Co-amoxiclav 1.2 g IV q8H **OR** Ampicillin/Sulbactam 3 g IV q6H	Cefazolin 2 g IV q8H + Metronidazole 500 mg IV q8H	• Duration: 7 - 10 days (subjected to clinical assessment) • With toxin production: Add Clindamycin 900 mg IV q8H for 7 days • If Necrotizing fasciitis is suspected, please start IV Clindamycin (refer below) • Surgical debridement is URGENT. • Antibiotics should be streamlined based on intraoperative culture results.
Severe	Piperacillin/Tazobactam 4.5 g IV q6H + Clindamycin* 900 mg IV q8H ***(if necrotising fasciitis is suspected or confirmed)***	**Penicillin Allergy:** Meropenem 1 g IV q8H + Clindamycin* 900 mg IV q8H ***(if necrotising fasciitis is suspected or confirmed)***	• Surgical debridement is URGENT. • Antibiotics should be streamlined based on intraoperative culture results. • Total duration: 7 – 10 days. Duration is suggested and can be adjusted depending on clinical progress. • IV antibiotics should be de-escalated to oral once clinical improvement occurs. • *Clindamycin is used to stop toxin production in infections due to *Staphylococcus aureus, Streptococcus pyogenes* and some anaerobic gram-positive organisms. Clindamycin

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			should be stopped once clinical improvement occurs.
Chronic, recurrent diabetic foot ulcer, no evidence of infection	No antibiotics, daily dressing, wound care Consider referral to UMMC Diabetic Care Unit.		
Necrotising fasciitis			
Monomicrobial *Streptococcus spp,* *Staphylococcus spp,* *Clostridium spp,* *Vibrio spp,* *Aeromonas hydrophila* **Polymicrobial** In diabetes/perianal abscess, post-operative wound infection (esp. abdominal wound), mixed	Piperacillin/Tazobactam 4.5 g IV q6H + Clindamycin* 900 mg IV q8H	**Penicillin allergy:** Meropenem 1 g IV q8H + Clindamycin* 900 mg IV q8H	• Surgical debridement is URGENT. • Please send intraoperative tissue cultures. • Antibiotics should be streamlined based on intraoperative culture results. • Total duration: 7 -10 days. Duration is suggested and can be adjusted depending on clinical progress. • IV antibiotics should be de-escalated to oral once clinical improvement occurs • *Clindamycin is used to stop toxin production in infections due to *Staphylococcus aureus, Streptococcus pyogenes* and

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
aerobes-anaerobes			some anaerobic gram-positive organisms • Clindamycin should be stopped once clinical improvement occurs or source control done (max duration 7 days). • Add Vancomycin if MRSA is suspected.
Animal/Human Bites Purulent: Polymicrobial, mixed aerobes and anaerobes Non-purulent wounds: *Staphylococcus spp., Streptococcus spp.*			
Mild	Co-amoxiclav 625 mg PO q8H	Cefuroxime 500 mg PO q12H + Metronidazole 400 mg PO q8H **OR** Co-trimoxazole 2 SS tablets (480 mg) PO q12H +	

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
		Metronidazole 400 mg PO q8H	
Moderate	Co-amoxiclav 1.2 g IV q8H **OR** Ampicillin/Sulbactam 3 g IV q6H	Cefuroxime 1.5 g IV q8H + Metronidazole 500 mg IV q8H	
Severe sepsis	Piperacillin/Tazobactam 4.5 g IV q6H	Meropenem 1 g IV q8H	
For monkey bite (only macaque family) *Herpes simiae* Herpes B virus Monkey B virus	As above + Post-exposure prophylaxis*		
Non-diabetic Ulcer			
Non-infected	Wound care, no antibiotics required.		

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Infected	For wound debridement or biopsy for HPE, cultures and mycobacterium cultures. To treat based on cultures.		
Herpes Zoster			
Immunocompetent	Aciclovir 800 mg PO five times daily for 7 days		• Antiviral treatment indicated for immunocompetent patients who present within 72 hours of onset of rash, and for all immunocompromised patients regardless of duration of rash. • Also treat pregnant women with the same dose for oral.
Immunocompromised patients with disseminated disease	Aciclovir 10 mg/kg IV q8H After clinical improvement, change to oral. Total duration: 7 days		
Herpes Simplex			

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Oral lesions, mouth, lips	Aciclovir 400 mg PO five times daily	Aciclovir 5% cream	
Leprosy *Mycobacterium leprae, Mycobacterium lepromatosis*			
<u>Paucibacillary:</u> **Tuberculoid leprosy (TT),** **Borderline tuberculoid (BT)**	Rifampicin 600 mg PO monthly + Dapsone 100 mg PO q24H*, for 6 months	Rifampicin 600 mg PO monthly + Clofazimine 300 mg PO monthly + Clofazimine 50 mg PO q24H + Dapsone 100 mg PO q24H	• Paucibacillary (PB) case: a case of leprosy with 1 to 5 skin lesions, without demonstrated presence of bacilli in a skin smear. • Multibacillary (MB) case: a case of leprosy with more than five skin lesions; or with nerve involvement (pure neuritis, or any number of skin lesions and neuritis); or with the demonstrated presence of bacilli in a slit-skin smear, irrespective of the number of skin lesions. • *To check G6PD status • **WHO guideline 2018 • *** Ofloxacin 400 mg is not available in UMMC, can be replaced with Levofloxacin 500 mg Tablet • ****Minocycline is not available in UMMC.
<u>Multibacillary:</u> **Mid-borderline (MB),** **Borderline lepromatous**	Rifampicin 600 mg PO monthly + Clofazimine 300 mg PO	Same regime for both PB (6 months duration) and MB (12 months duration) **	

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
(BL), **Lepromatous leprosy (LL)**	monthly + Clofazimine 50 mg PO q24H + Dapsone 100 mg PO q24H*, for 12 months		
Rifampicin-resistant Leprosy**	Ofloxacin 400 mg*** PO q24H + Minocycline 100 mg**** PO q24H + Clofazimine 50 mg PO q24H, for first 6 months **Then,** Ofloxacin 400 mg*** PO	Ofloxacin 400 mg*** PO q24H + Clarithromycin 500 mg PO q24H + Clofazimine 50 mg PO q24H, for first 6 months **Then,** Ofloxacin*** 400 mg PO	

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	q24H **OR** Minocycline 100 mg**** PO q24H + Clofazimine 50 mg PO q24H, for next 18 months	q24H + Clofazimine 50 mg PO q24H, for next 18 months	
Rifampicin and Ofloxacin-resistant Leprosy**	Clarithromycin 500 mg PO q24H + Minocycline 100 mg**** PO q24H + Clofazimine 50 mg PO q24H, for first 6 months	Clarithromycin 500 mg PO q24H **OR** Minocycline 100 mg**** PO q24H + Clofazimine 50 mg PO q24H, for next 18 months	
Acne Vulgaris			

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Oral antibiotics should be used for the shortest possible duration to minimise risk of adverse effects and development of bacterial resistance. Typical course is 3 months.			
	Tetracycline 1 g PO daily in divided doses. Reduce gradually to 125-500 mg daily once improvement is noted. **OR** Doxycycline 100 mg PO q12H	Erythromycin Ethyl Succinate 400-800 mg PO q12H, followed by 400-800 mg PO q24H **OR** Azithromycin 500 mg PO q24H for 4 consecutive days per month for 3 consecutive months **OR** Azithromycin 500 mg PO q24H for 3 days, followed by 500 mg once a week for 10 weeks **OR** Azithromycin 500 mg PO q24H for 3 consecutive days each week in month 1, followed by 500 mg PO q24H for 2 consecutive days each	

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
		week in month 2, then 500 mg PO q24H for 1 days each week in month 3	
Rosacea			
	Doxycycline 100 mg PO q12-24H		
Nail Infection			
Onychomycosis Dermatophytes (*Trichophyton rubrum* and *T. mentagrophytes*) Candida, moulds (rare)	Terbinafine* is the first-line agent ***Fingernail onychomycosis:*** 250 mg PO q24H for 6 weeks ***Toenail onychomycosis:*** 250 mg PO q24H for 12 weeks	Itraconazole can be administered as either pulse or continuous therapy. **Pulse therapy** ***Fingernail onychomycosis***: 200 mg PO q12H for one week per month for two months ***Toenail onychomycosis:*** 200 mg PO q12H for one week per month for	• *Terbinafine is not available in UMMC.

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	Alternative	
		three months **Continuous therapy** ***Fingernail onychomycosis:*** 200 mg PO q24H for 6 weeks ***Toenail onychomycosis:*** 200 mg PO q24H for 12 weeks **OR** Fluconazole 150-450 mg PO weekly for 3-12 months	
Fungal Skin Infection			
Tinea corporis/cruris *Trichophyton rubrum* *T. mentagrophytes*	**Topical:** Terbinafine 1% cream* apply q12H for 1 week	**Topical:** Miconazole 2% cream	• *Not available in UMMC Formulary

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Epidermophyton floccosum	**OR** Ketoconazole 2% cream		
	Oral **(*Alternative for patients with extensive skin involvement and patients who fail topical therapy*):** Itraconazole 100-200 mg PO daily for 1 week **OR** Fluconazole 150-300 mg PO daily for 4-6 weeks **OR** Fluconazole 150-200 mg PO weekly for 2-4 weeks **OR** Griseofulvin (micronized) 500 mg PO daily for 2-4 weeks		

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Tinea capitis Requires systemic treatment	Griseofulvin (micronized) 500 mg PO daily for 2-4 weeks **OR** Itraconazole 100-200 mg PO q24H for 2-4 weeks **OR** Fluconazole 200-400 mg PO q24H for 3-6 weeks		
Tinea barbae Requires systemic treatment	Griseofulvin (micronized) 500 mg PO q24H for 6 weeks **OR** Itraconazole 200 mg PO q24H for 2-4 weeks **OR** Fluconazole 200 mg PO q24H for 4-6 weeks		
Tinea pedis/manuum	Itraconazole 200 mg PO		

CHAPTER 3.10	SKIN AND SOFT TISSUE INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	q12H for 1 week **OR** Itraconazole 200 mg PO q24H for 2 weeks **OR** Fluconazole 150 mg PO weekly for 3-4 weeks		
Intertrigo			
Candida spp	Topical Miconazole **OR** Topical Ketoconazole **OR** Topical Clotrimazole Powder	**For refractory cases:** Fluconazole 100 mg PO q24H	• General management: weight loss, control of diabetes, drying agents